

AN2DL - First Challenge Report

DarkHorse

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1 Introduction

This project addresses the *multiclass time-series classification*, aiming to predict subject pain levels from sequential sensor data. The objective is to develop a model that generalizes robustly to unseen sequences while managing noisy inputs. Our structured approach involved two main phases:

- 1. Data Analysis and Feature Engineering:** Analyzing, cleaning, and transforming the dataset to maximize feature predictive power.
- 2. Model Architecture Definition:** Experimenting with various neural networks to capture both local spatial features and long-range temporal dependencies.

2 Problem Analysis

The dataset consists of motion sensor recordings from various pirates. Each 160-time step sequence tracks the position of 31 body joints and 4 pain survey metrics. Samples include categorical meta-features (number of legs, hands, eyes). Our Exploratory Data Analysis (EDA) focuses on:

Analysis on Joints: We analyzed variance, correlation and Mutual Information (MI) for each joint feature and identified the presence of outlier features.

Temporal Periodicity: Analysis of the joint position data revealed that, despite the monotonic time

index, the joints exhibit inherent periodicity and cyclicity.

Class Imbalance: The dataset is heavily skewed. The majority class `no_pain` accounts for 77% of samples, leaving 14% for `low_pain` and 8% for `high_pain`. This disparity creates a strong bias, leading to performance degradation when predicting the underrepresented classes.

Autocorrelation Analysis: We assessed the motion signal's "memory" via autocorrelation analysis using several joints. It consistently revealed that meaningful correlations between time steps decay after a specific lag count. This quantitative finding was crucial for determining the optimal sliding window size needed to efficiently capture relevant dynamics while excluding excessive temporal noise.

3 Method

3.1 Data Preprocessing

Feature Filtering: Features with near-zero variance, low mutual information, or high correlation were pruned. This step reduced dimensions from 31 to ~ 20 , improving model stability and computational efficiency.[2]

Temporal Feature Engineering: Each frame was augmented with cyclical time encodings ($t_{\sin} = \sin(\frac{2\pi t}{T})$, $t_{\cos} = \cos(\frac{2\pi t}{T})$) allowing the model to

exploit periodic structures that raw timestamps fail to capture.[1]

Normalization: To prevent data leakage, all feature normalization was performed using statistics computed exclusively from the training fold. Joint features were standardized using **Z-score normalization**: $x' = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$. Pain surveys were rescaled to $[0, 1]$, while metadata was binary-encoded.

Class Imbalance Mitigation: To address class imbalance, we applied **oversampling** with light augmentation (Gaussian jitter on joint coordinates) to minority classes.

Sliding Window Segmentation: Windows of 40 frames and a stride of 20 were extracted to capture local temporal dynamics while augmenting the training set from 661 to $\sim 3,305$ samples.

3.2 Model Architecture

Our classifier comprises a **hybrid** deep neural architecture integrating convolutional, recurrent, and attention mechanisms to capture both *local motion dynamics* and *global temporal contexts*.

CNN Feature Extractor: Two 1D convolutional layers process each temporal window:

$$\text{Conv1D} \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPool} \quad (1)$$

The CNN learns short-term patterns and serves as a noise reduction mechanism, improving signal clarity for subsequent recurrent layers.

Bidirectional LSTM Encoder: The extracted convolutional features are processed by a bidirectional LSTM, which captures long-range temporal dependencies across the entire window by encoding context from both past and future time steps.

Attention Mechanism: A soft attention layer adaptively weights temporal segments, prioritizing informative frames while suppressing background noise. Due to its expressive power, careful regularization was essential to prevent overfitting.[3]

Classification Head: The attention vector is then passed through a two-layer MLP with dropout regularization, producing the final class probability.

3.3 Training Procedure

Stratified K-fold Cross-Validation: To ensure robust generalization, we used 7-fold stratified cross-validation with subject-level splitting. This prevents data leakage by ensuring all temporal frames from the same pirate remain within a single fold, while stratification preserves the class distribution across all splits.

Optimization and Regularization: The model was trained using **cross-entropy loss** with **label smoothing** ($\epsilon = 0.1$) to prevent overconfident predictions. Optimization used **AdamW** coupled with **ReduceLROnPlateau scheduler** for adaptive learning rate decay.

Regularization is implemented through:

- Dropout rate in CNN and classifier layers
- L2 regularization via weight decay in the optimizer
- Gradient clipping (max norm = 0.8) to prevent exploding gradients
- Early stopping with patience of 50 epochs based on validation F1 score

Inference and Aggregation: At inference time, predictions were aggregated per subject by averaging class probabilities:

$$\hat{y}_{\text{pirate}} = \frac{1}{N_{\text{windows}}} \sum_{i=1}^{N_{\text{windows}}} \hat{y}_i \quad (2)$$

The final predicted class is determined via $\arg \max(\hat{y}_{\text{pirate}})$.

Ensemble Strategy: Final test predictions were obtained by averaging predicted probabilities across all 7 folds, creating a robust ensemble that reduces variance and improves generalization beyond single-fold models.

4 Experiments

4.1 Experimental setup

We first compared several sequence models (LSTM, GRU and simple autoencoder-classifier) on a fixed train/validation split, identifying the BiLSTM with Min-Max normalization as the best initial option.

In later stages we switched to a 7-fold stratified cross-validation with **hybrid 1D CNN + BiLSTM + attention** architecture and temporal windowing. We refined the data processing pipeline by augmenting minority classes, pruning joint features, and moving to z-score normalization for joint features.

4.2 Quantitative results

Table 1 reports the validation weighted F1 achieved by the three main configurations described in Section 4.1.

Table 1: Weighted F1 of each model

Model	Weighted F1
Baseline BiLSTM	0.9198
CNN-LSTM-Attention (no pruning)	0.9488
CNN-LSTM-Attention (final, pruned)	0.9696

Figure 1 shows the validation weighted F1 for each fold of the best model, indicating stable performance across the seven cross-validation splits. Figure 2 shows the confusion matrix.

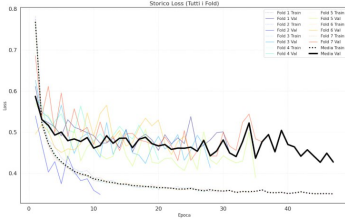


Figure 1: Validation weighted F1-score for each cross-validation fold of the final model.

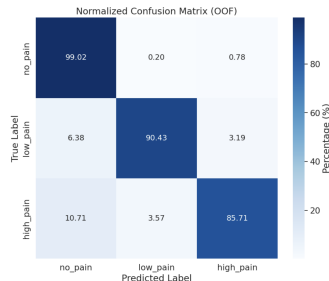


Figure 2: Confusion matrix.

5 Results

The hybrid model successfully met the primary objectives of high predictive performance and robust generalization, showing substantial improvement over the baseline.

Validation Performance: As shown in Table 1, the final model (M3) achieved a weighted F1 score of 0.9696 on the validation set, significantly outperforming the baseline and the intermediate configuration.

Stability and Generalization: The model’s reliability is confirmed by the minimal variance observed across the 7 cross-validation folds Figure 1. Furthermore, the low generalization gap between the Out-of-Fold (OOF) validation scores and the final test performance suggests that the applied regularization and feature engineering strategies successfully mitigated overfitting.

6 Discussion

The proposed CNN-LSTM-Attention model shows strong, performance across folds, confirming the efficacy of the combination of windowing, feature pruning, augmentation and z-score normalisation. However, a key weakness is the overall **complexity of the pipeline**: the model is relatively heavy and heavily regularised, which suggests that a simpler model might achieve comparable performance under a similar training and evaluation setup.

7 Conclusions

We successfully developed a hybrid CNN-LSTM-Attention model for multi-class time-series pain recognition, supported by a pipeline (feature pruning, temporal windowing, data augmentation, and careful cross-validation). The **main contribution** is the combining local pattern extraction, long-term temporal modelling and attention-based aggregation within a robust training framework that achieves strong and stable performance. Despite its complexity, future work should first explore whether simpler, more **interpretable** architectures. Further improvements include more advanced augmentation, adaptive window sizes, and the adoption of Transformer-based models.

References

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